① local search

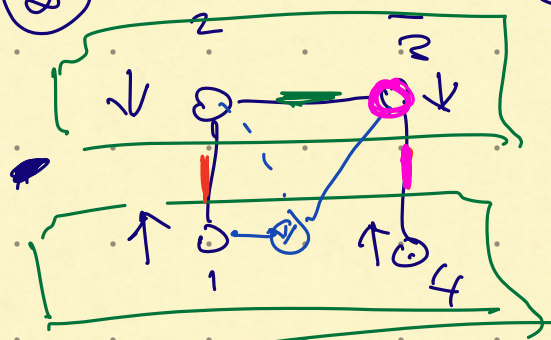
ⓐ Requirement: find $\max_{\sigma} C(\sigma)$.

$$C(\sigma) = \frac{1}{4} \sum_{u,v} A_{uv}(1 - \sigma_u \sigma_v)$$

ⓑ Operation:

start with some σ

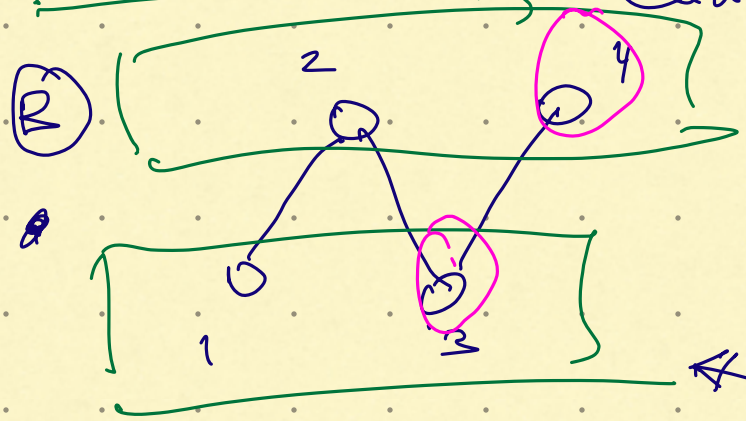
② $\sigma = (\sigma_1, \sigma_2, \cdots, \sigma_N)$



Cut = 2.

Cannot be improved.

③



Cut = 3

maximum.

BPG!

③ k-opt local search.

2-opt local search.

Expensive:

1-opt. N options

2-opt $\frac{N(N-1)}{k}$

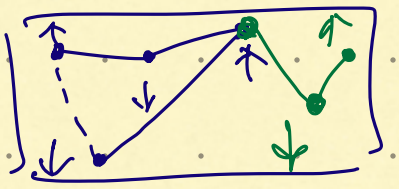
k-opt C_N^k

$C_N^k = \frac{N!}{k!(N-k)!} \sim \exp(\cdot k)$

1-opt is parallelizable by relaxation.

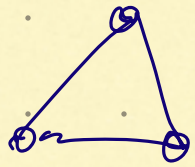
2-opt. ?

Bipartite graphs are trivial

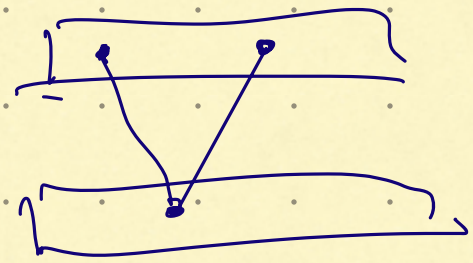


This works!

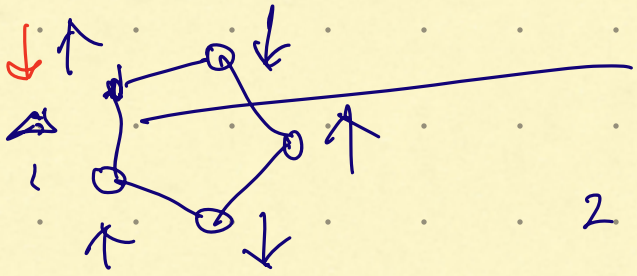
Att. There are odd cycles in BP graphs.

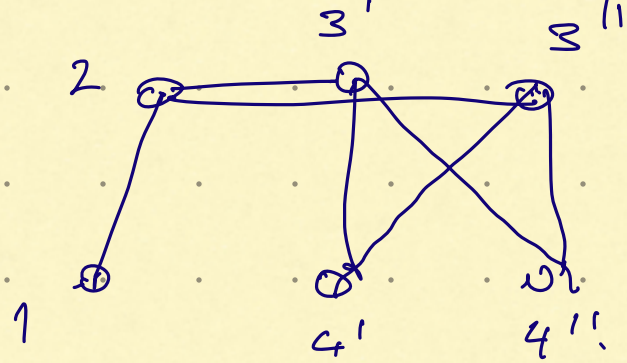


$\triangle$ C_3 odd.

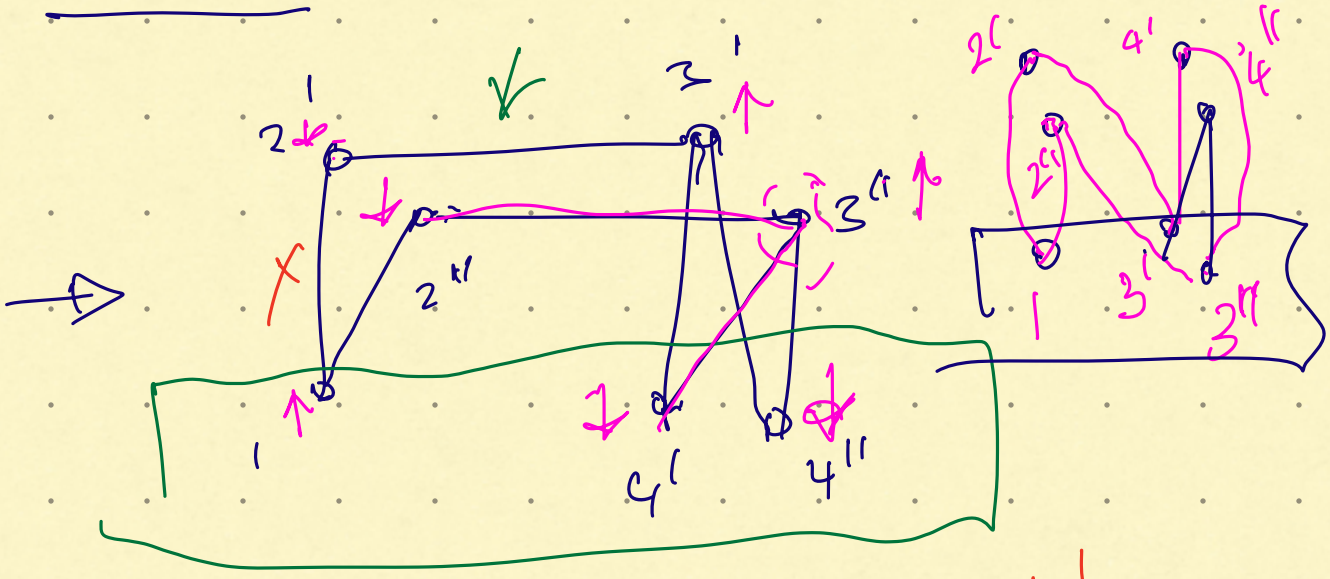


Require even edges to come back.





Cut = 4. Max-cut = 6.



None of the pairs will work!

2-opt LS also fails!

1-opt. 70%.

2-opt 15%.

(15) → 1 hour.

(4) Uphill climbing algorithms.

Simulated annealing.

← Metropolis algorithm.

$\sigma \rightarrow \sigma_n \rightarrow -\sigma_n$ — improve: Do this. / with probability

uphill. → $\sim \exp\left(\frac{\Delta C}{T}\right)$

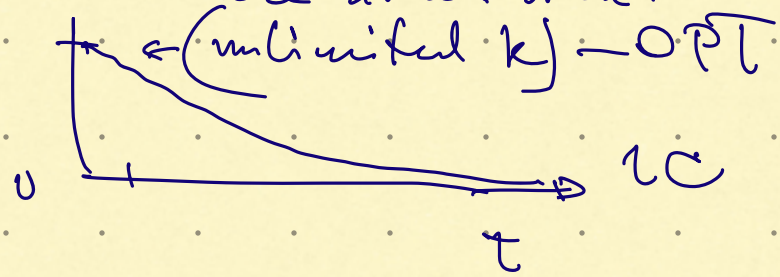
$T \sim 0$ almost no transition will happen.

$\hookrightarrow$ LC (Right LS)

$T \to 0$.

Long chains of transitions.

T are admitted.



Schedule. Good schedule is a question.

$$\dot{\xi}_m = \sum_n A_{mn} \xi_n + \gamma(t) \sqrt{T}$$

$\xi_m \in [0,1]$.

γ noise!

(8) SA still fails!

Ideal Schedule should be exp.-long.

(9) Max-cut is NP-hard.

depends on P vs NP.

APX-hard. approximation are NP-hard.

ρ - approximation ratio.

Algo: $G \rightarrow T$

$$\rho = \frac{\text{Worst-case } C_G(T)}{C_G}$$

$\rho_{LS} = \frac{1}{2}$. when $N \to \infty$

$\rho_{Random} = \frac{1}{2}$

To outperform this time $\sim e^N$.

$\rho_{k\text{-opt}} = \frac{1}{2}$.

$\rho_{SA\text{-non-exp}} = \frac{1}{2}$

$c_{SDP} \sim 0.878.$

Unique Games Conjecture -